

EC 441 — Practice Midterm 1 (Week 6 Assignment)

Name: _____ Student ID: _____

Instructions

Show all work. Leave answers clearly in the space provided.

Part I — Multiple Choice [20 pts]

Q1. Which delay depends most on network congestion?

A. Transmission B. Propagation C. Processing D. Queueing

Answer: _____

Q2. Shannon entropy is best interpreted as:

A. Max transmission rate B. Avg information per symbol C. Total symbols D. Error probability

Answer: _____

Q3. Ethernet switches learn MAC addresses by:

A. ARP B. IP headers C. Source MAC + port D. Config tables

Answer: _____

Q4. Manchester encoding is used because it:

A. Doubles data rate B. Reduces noise C. Enables clock recovery D. Eliminates collisions

Answer: _____

Q5. In CSMA/CD, collisions occur when:

A. Frames large B. Simultaneous transmission C. High attenuation D. CRC fails

Answer: _____

Q6. Increasing SNR affects capacity how?

A. Decreases B. Linear increase C. Log increase D. No effect

Answer: _____

Q7. Lowest attenuation medium:

A. Twisted pair B. Coax C. Fiber D. Wireless

Answer: _____

Q8. Pure ALOHA max efficiency:

A. 1 B. 0.5 C. 0.37 D. 0.18

Answer: _____

Q9. CRC works using:

A. Parity B. Polynomial division C. Encryption D. Checksums

Answer: _____

Q10. Routing occurs at:

A. Application B. Transport C. Network D. Link

Answer: _____

Part II — Short Answer [30 pts]

SA1. Delay Analysis: 1 MB file, $R=10$ Mb/s, $d=1000$ km, $v=2 \times 10^8$ m/s.

(a) Transmission delay (b) Propagation delay

Answer:

SA2. Shannon Capacity: $B=1$ MHz, $SNR=30$ dB.

(a) Convert SNR (b) Capacity

Answer:

SA3. Entropy: $p(A)=0.5$, $p(B)=0.25$, $p(C)=0.25$. Compute $H(X)$.

Answer:

SA4. Attenuation: 0.2 dB/km over 50 km.

(a) Loss (b) Output power

Answer:

SA5. CRC Concept: Explain detection and response.

Answer:

Part III — Problem Solving [50 pts]

P1. End-to-End Delay: 5 MB file, 3 links, $R=10$ Mb/s, $d=1500$ km.

Answer:

P2. Shannon vs Real: 20 Mb/s vs 50 Mb/s capacity.

Answer:

P3. ALOHA: equation, max efficiency, explanation.

Answer:

P4. Ethernet Switching scenario.

Answer:

P5. 16-PAM at 1 Msymbol/s: bits/symbol, bit rate, tradeoffs.

Answer: